Towards Automated Sleep Analysis: Stage Classification and Apnea Prediction

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Background

Background & Motivation:

- ~70 million U.S. adults have sleep disorders; Obstructive Sleep Apnea (OSA) is among the most prevalent and underdiagnosed.
- Poor sleep quality independently elevates risk of chronic disease, cognitive decline, and cardiovascular conditions.

Goal 1: Build a model to predict sleep stage real-time.

Goal 2: Build a model to predict apnea events 10 seconds in advance

Dataset

We used DREAMT dataset from Physionet

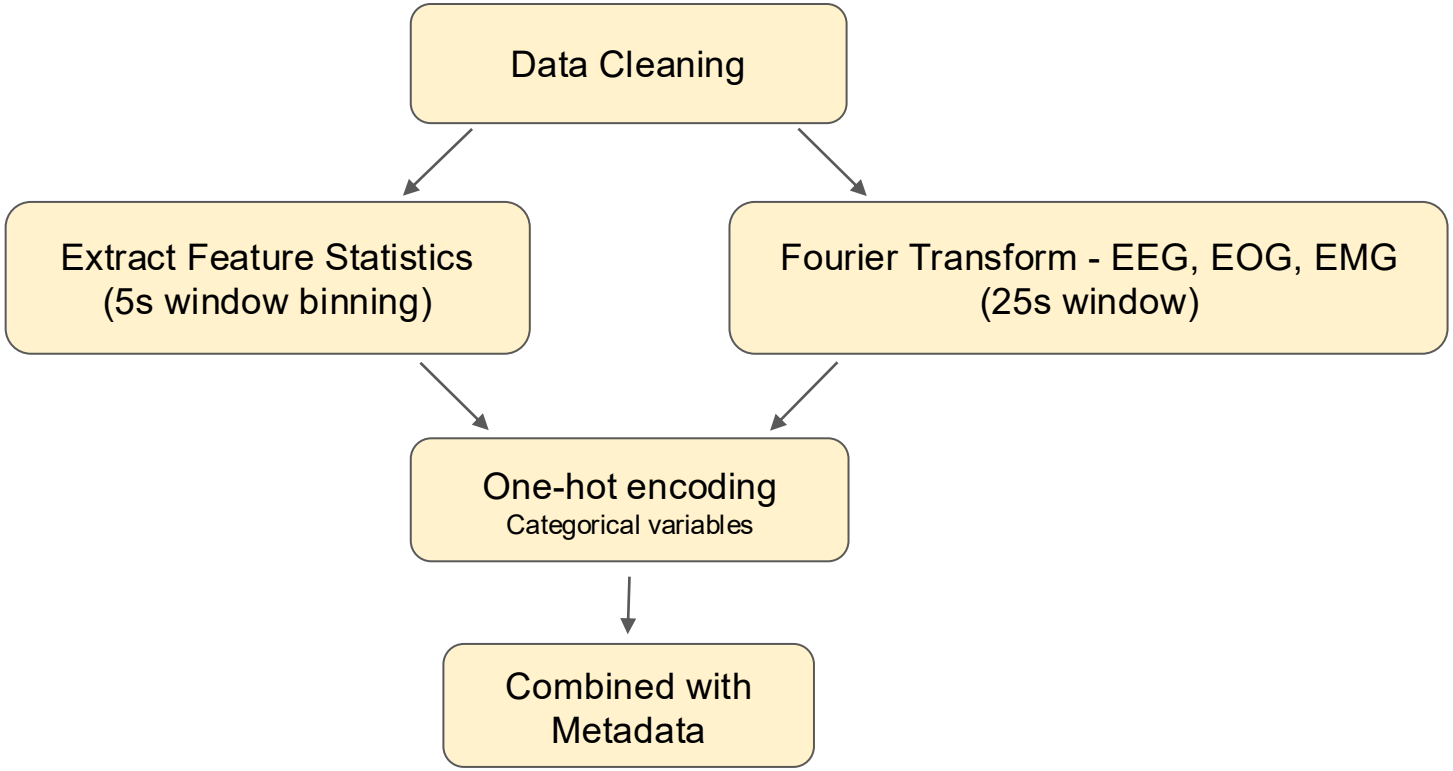
- Collected from 100 participants(May–Sept 2022).
- Participants wore an smartwatch throughout an overnight Polysomnography study.

Data:

- Metadata for 100 subjects, e.g. Age, bmi, medical history
- Time-series signal (100 Hz), e.g. EEG, EOG, EMG, heart rate, respiratory signals
- Sleep stage (manually annotated): Wake, N1, N2, N3, REM
- Apnea event (manually annotated)

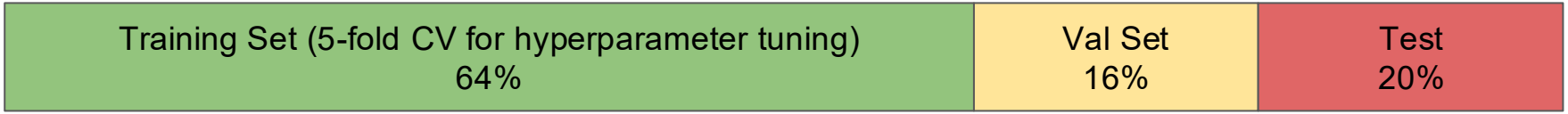
Dataset available at: https://physionet.org/content/dreamt/2.1.0/.

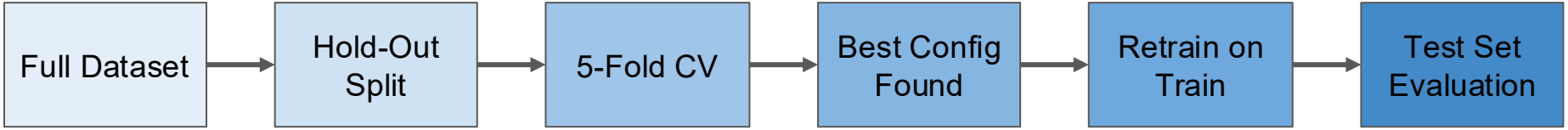
Preprocessing



End-to-End Experimental Pipeline

100 subjects are divided into three non-overlapping partitions





Goal 1: Sleep stage prediction

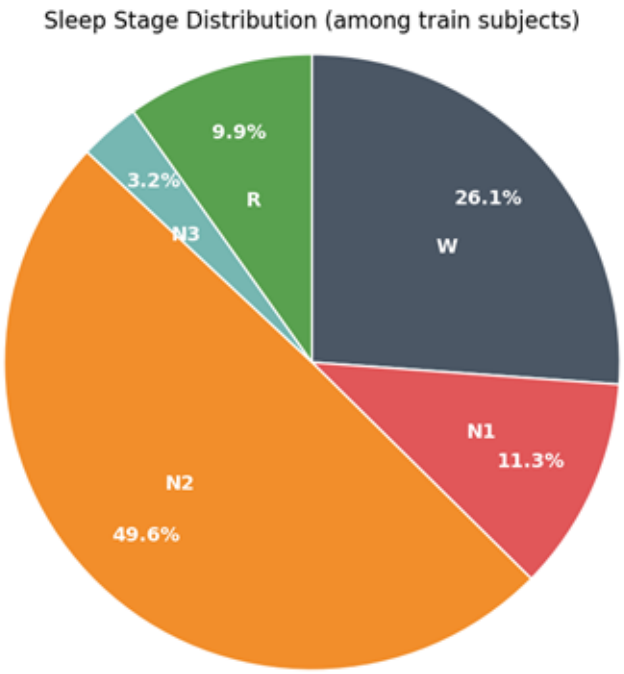
Objective: **classify 5 sleep stages (Wake, N1, N2, N3, REM)** at each time step using PSG signals, wearable device signals and subject metadata

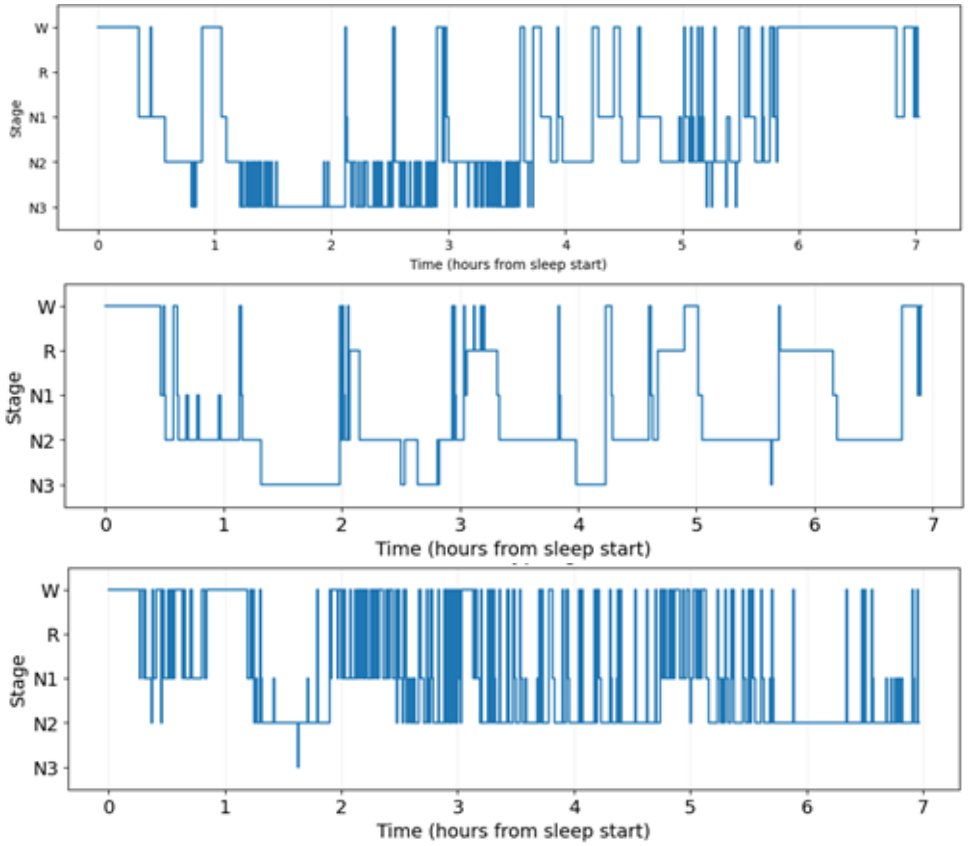
Metric: **Macro F1** – equal importance across all stages

Challenges:

- Large subject variability
- Class imbalance and underrepresented transient stages

EDA


Sleep Stage Distribution (among train subjects)
9.9%
R
3.2%
N3
26.1%
W
N1
11.3%
N2
49.6%


W
R
N1
N2
N3
Stage
0
1
2
3
4
5
6
7
Time (hours from sleep start)

Models - training procedures

Preprocessed raw data
Dataset loader
train/val/test split

Feature Engineering:
- reduce numerical precision for less noise
- EEG features: relative band-power + cross-band ratios
- Lag temporal features for dominant signals

Tabular Models (4Fold CV)
hyperparameter search + class weights

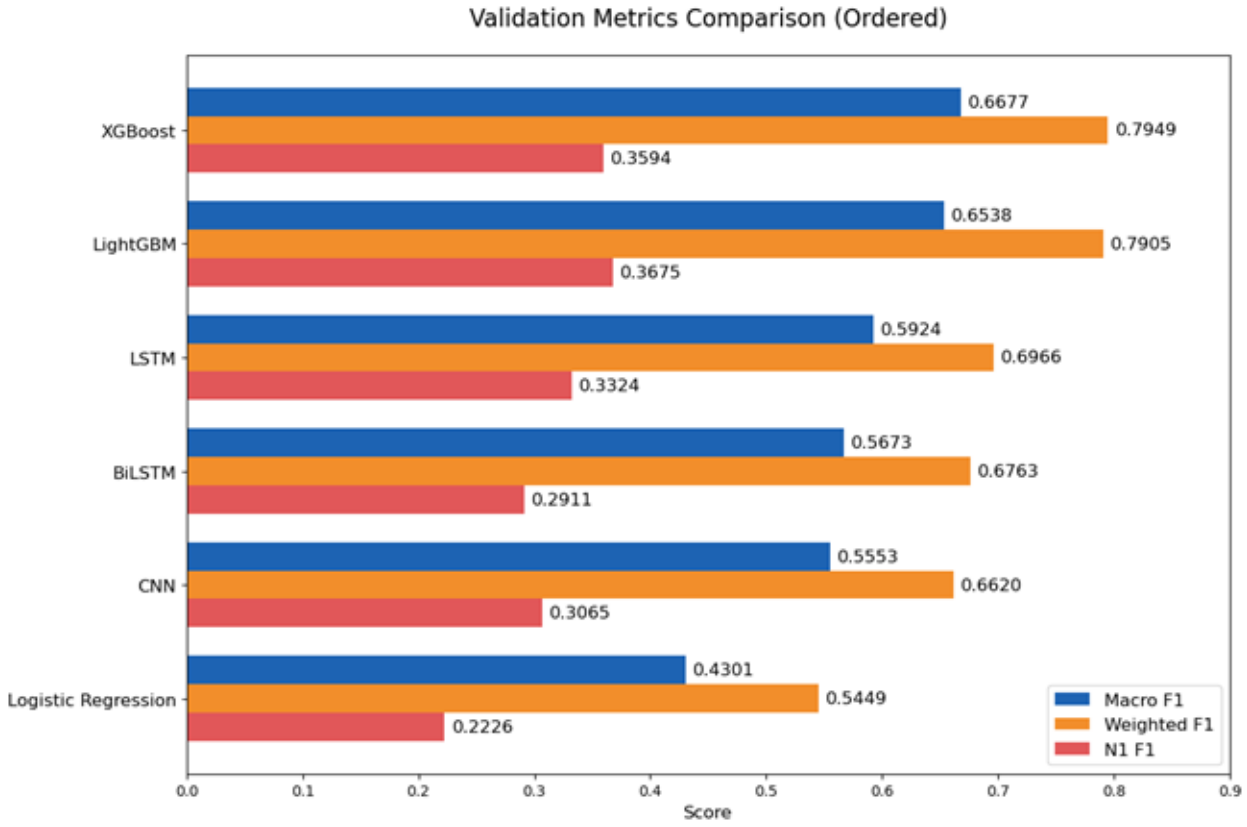
- Logistic Regression
- XGBoost
- LightGBM

Standard scaling
Load dataset into fixed-length sequences

Training sequence models with iterative tunning and early stopping

- BiLSTM
- LSTM
- CNN

Validation Metrics

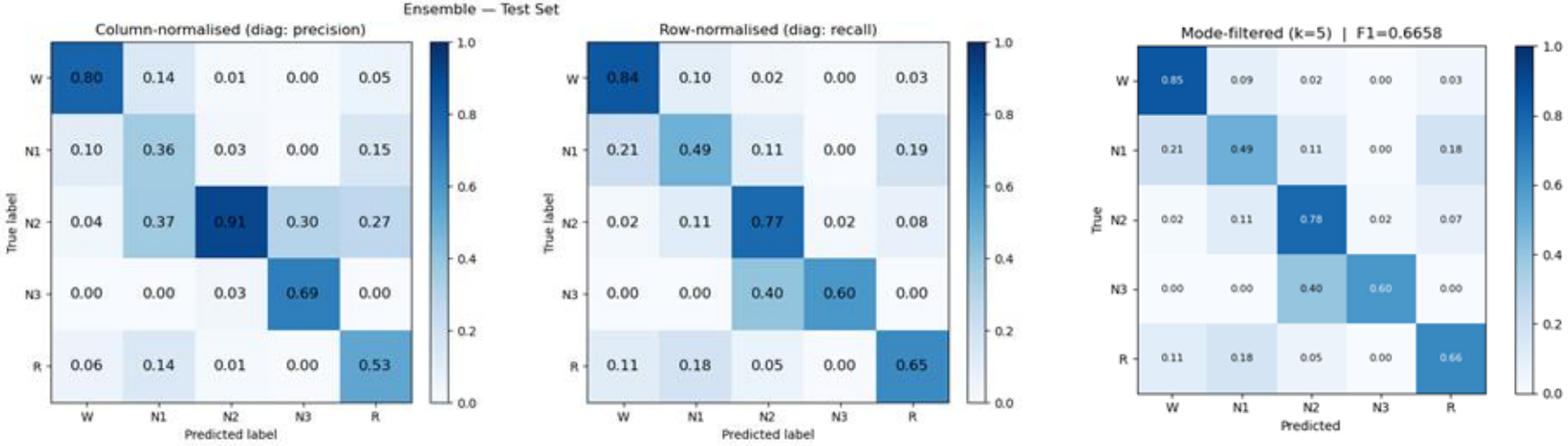


Model	Best Parameter
xgb	n_estimators = 630 max_depth = 6 learning_rate = 0.02685
lgb	n_estimators=507 learning_rate=0.0288 num_leaves=55 max_depth=6
LSTM	seq_len=30 lr=0.0005 hidden_dim=32
BiLSTM	seq_len=10 lr=0.001 hidden_dim=32
CNN	seq_len=30 lr=0.0005 channels=[32,64,64]
lgr	regularization=100

Test: Ensemble xgb + lgb

```
y_prob_xgb = xgb_model.predict_proba(X_test)
y_prob_lgb = lgb_model.predict_proba(X_test)
y_pred_ens = np.argmax(y_prob_xgb * 0.5 + y_prob_lgb * 0.5, axis=1)
```

```
kernel = 5
window = y[i-kernel//2 : i + kernel//2]
y[i] = max(Counter(window).values())
```

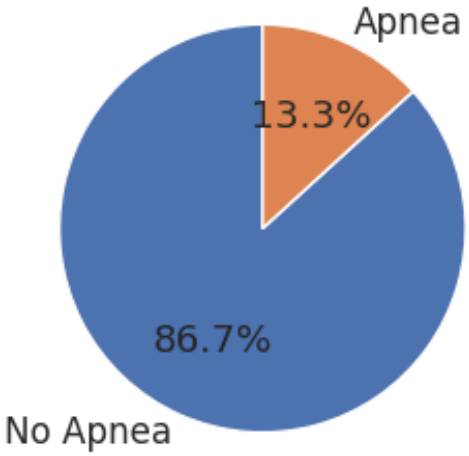


	XGBoost	LightGBM	Ensemble 50/50	Ens + mode5
Macro F1	0.6565	0.6579	0.6593	0.6658

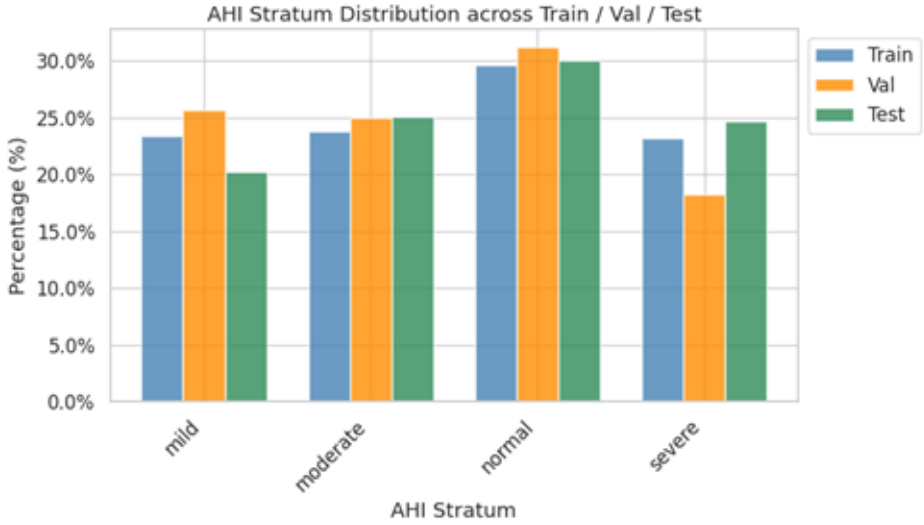
Goal 2: Future apnea event prediction

EDA

Class Imbalance



Stratified split by apnea severity

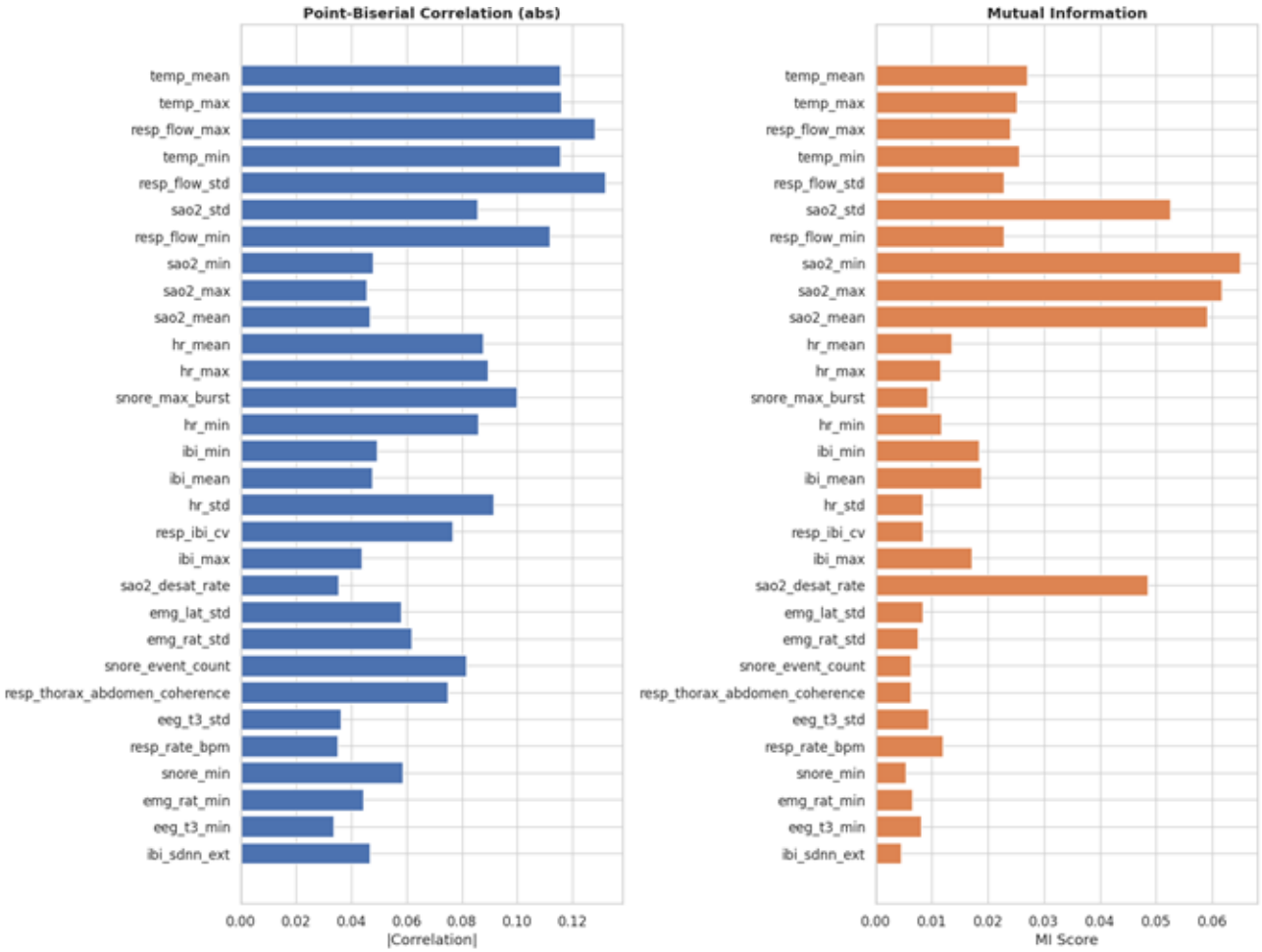


Apnea: no apnea ~ 1:7
Apnea frequency: train set 0.133, validation set 0.097, test set 0.127

Feature rank

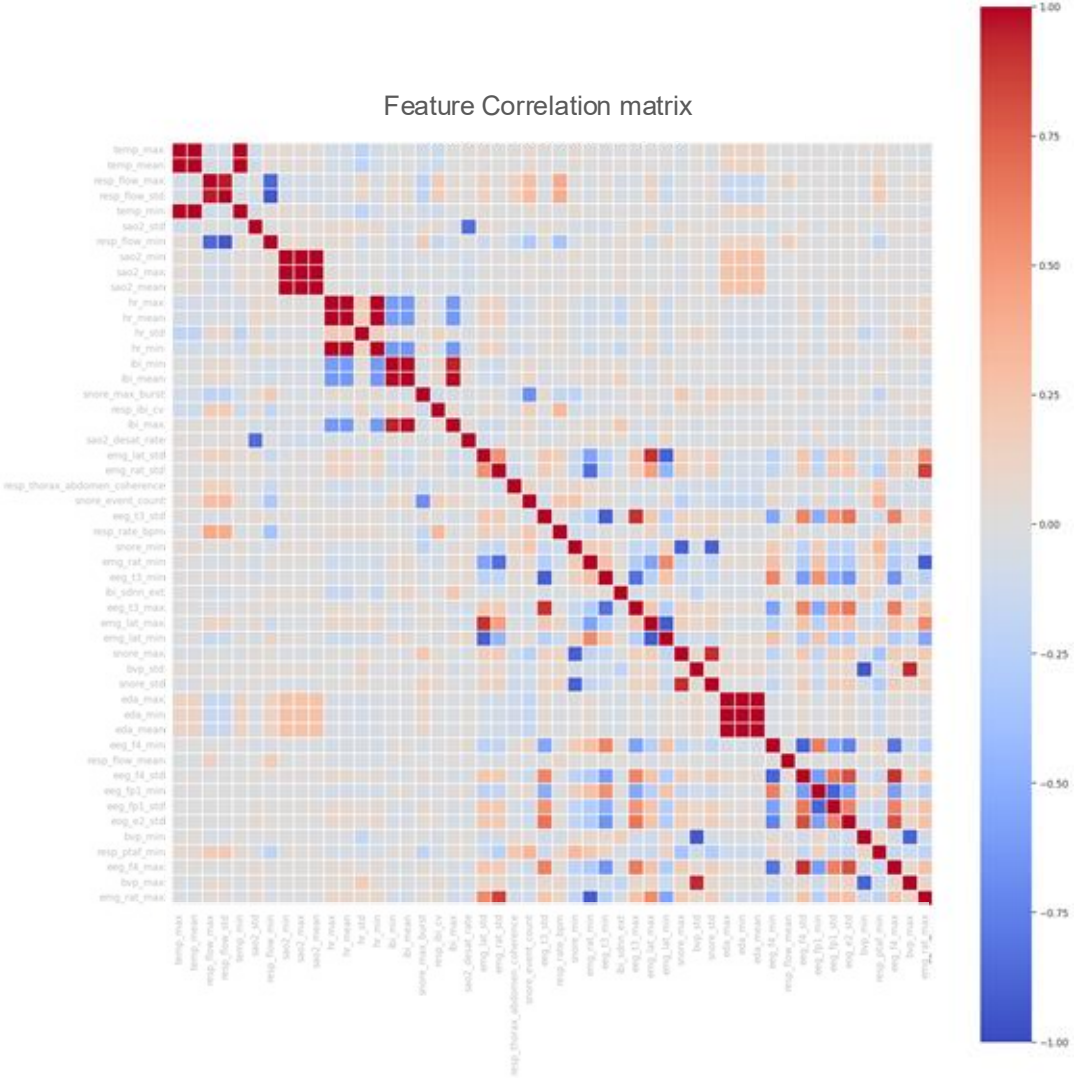
Rank features based on their point-biserial correlation and mutual information with target

Top 30 Features

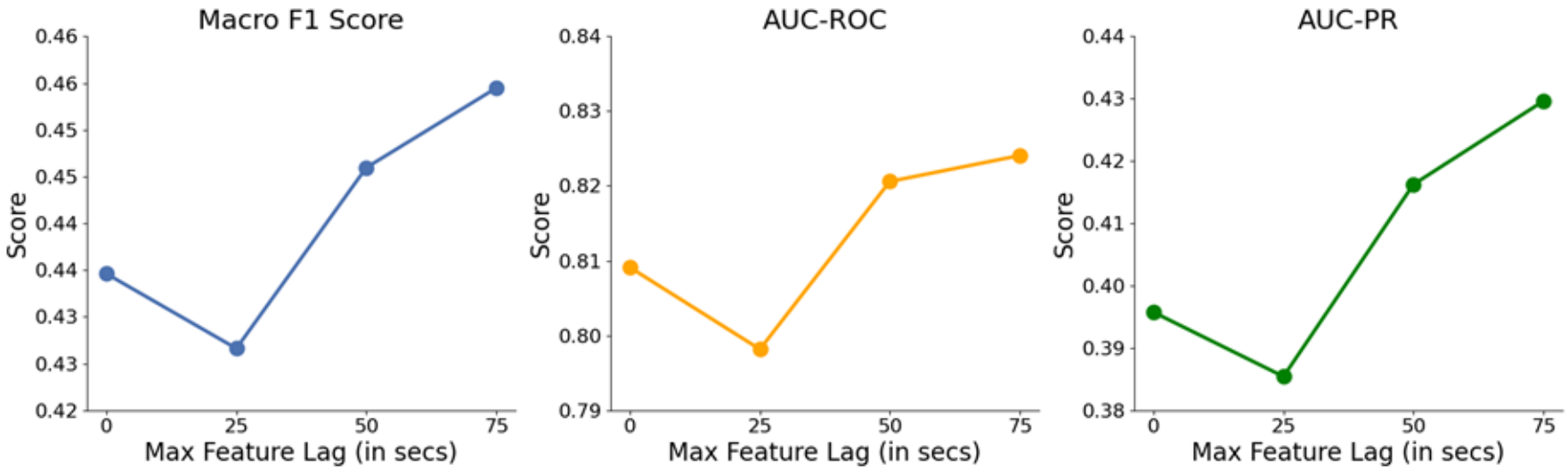


Feature removal

- Examine feature correlation → remove feature pairs with correlation > 0.9
- Examine feature variance → drop near-zero variance features with low ranking
- Result :127 → 67 time-series features



LightGBM with various feature lag



75s feature lag is the best

Final test metrics

AUC-ROC=0.786, AUC-PR=0.467, Accuracy= 0.88

	precision	recall	F1
No apnea	0.93	0.94	0.93
apnea	0.54	0.50	0.52
Macro avg	0.73	0.72	0.72
Weighted avg	0.88	0.88	0.88

Discussion

Sleep stage:

- The main limitation of real-time sleep stage prediction comes from **class imbalance and ambiguous transitions between stages**, rather than model complexity. Stage boundaries are inherently noisy, even with advanced PSG measurements
- Future direction: **improve per-stage discrimination,** and combine models strategically for a system reliable overall

Apnea event:

- 75s lag of top features yielded the strongest performance of LightGBM, which suggested that **extended temporal context is critical** for anticipating breathing disruptions.
- Future direction: apply LSTM or Transformer to include longer context